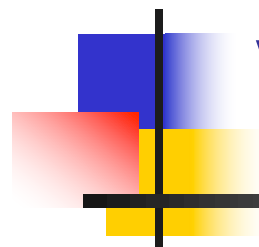
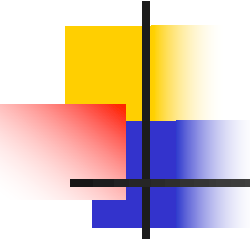


Advanced topics

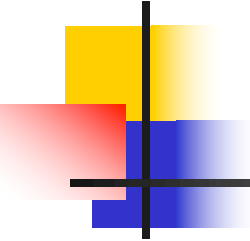


Validating models



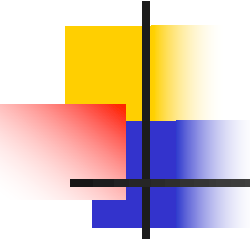
Cross-validation

- Cross-validation was proposed by Kurtz (1948-simple cross-validation) , extended by Mosier (1951-double cross-validation) and by Krus and Fuller (1982- multicross-validation).
- The original objective of cross-validation is to verify replicability of results.
- Similarly with hypothesis testing, the goal is to find out if the result is replicable or just a matter of random.



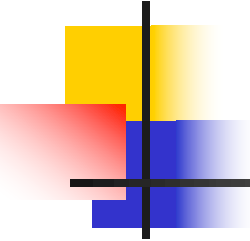
Cross-validation

- For a more realistic estimate of the prediction error, we need a test sample different from the training sample.
- One way is to split the observation **z** into two sets, one for training and the other one for testing.
- Using a part of the available observation to fit the model, and the part to test in the computation of predication error is known as the cross-validation.



Cross-validation

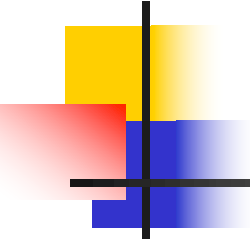
- **Simple cross-validation.** Divide the set z into two groups, one for training and the other one for testing.
 - The parameters are estimated on the training data set.
 - The cross-validation is the prediction error computed using the test sample.
- **Double cross-validation.** Models are generated for both sub-samples, and then both equations are used to generate cross-validation.



K-Fold cross validation

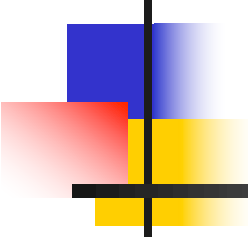
1. Split z into K equal subsets z_k
2. Do K times:
 - Estimate the model (k) on $z(k) = \{z_1, \dots, z_{k-1}, z_{k+1}, \dots, z_K\}$
 - Compute the prediction error $PE(k)$ between the test sample z_k and the predicted model by (k).
3. Compute the average of those K prediction errors as the overall estimate of the prediction error

$$\frac{1}{K} \sum_{j=1}^K PE(j)$$

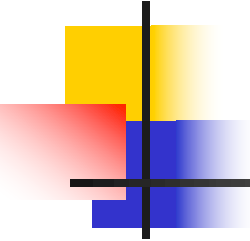


Cross validation and parameter selection

- We just show how to perform model selection (i.e. choice of the number of parameters p in the regression model).
- Cross-validation can also be used for parameter estimation by choosing the parameter value which minimises the prediction error.
- The cross-validation is computed using subsamples of z .
 - An alternative consists in considering bootstrap samples.

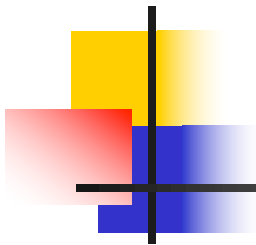


Early Effort Estimation of Massive Maintenance Processes



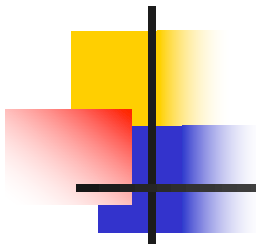
Case study

- Y2K massive maintenance project
- About 40,000 software components
- System decomposed into loosely-coupled, cohesive components, called WorkPackets (WP)
- Phases of the maintenance process:
 - Inventory and Assessment ← metrics extraction
 - Analysis, Design, Implementation, Testing, Delivery-Installation



The Model

- Correlation analysis performed between WP maintenance effort and
 - # of components in the WP (SC),
 - Mean value of metrics extracted from a given percentage of WP components (LOC, Cycl. Compl., Halstead volume, # of control variables, # of logical branches not used)
- The model should take into account, at least:
 - The average value of one dimensional metric
 - The average value of a structural metric
 - The number of components in the WP



The Model

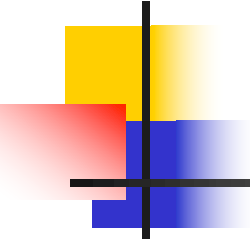
$$Eff = (\alpha_1 \cdot SC + \alpha_2) \cdot \log(\overline{LOC}) + (\alpha_3 \cdot SC + \alpha_4) \cdot \overline{CYC} + \alpha_5 \cdot SC$$

- Model parameters:

α_1	α_2	α_3	α_4	α_5
4.92	49.8	0.0741	4.11	39.98

- Significance statistics:

R²	0.9	Adjusted R²	0.85
F-statistic	16.34	p-value	0.000278

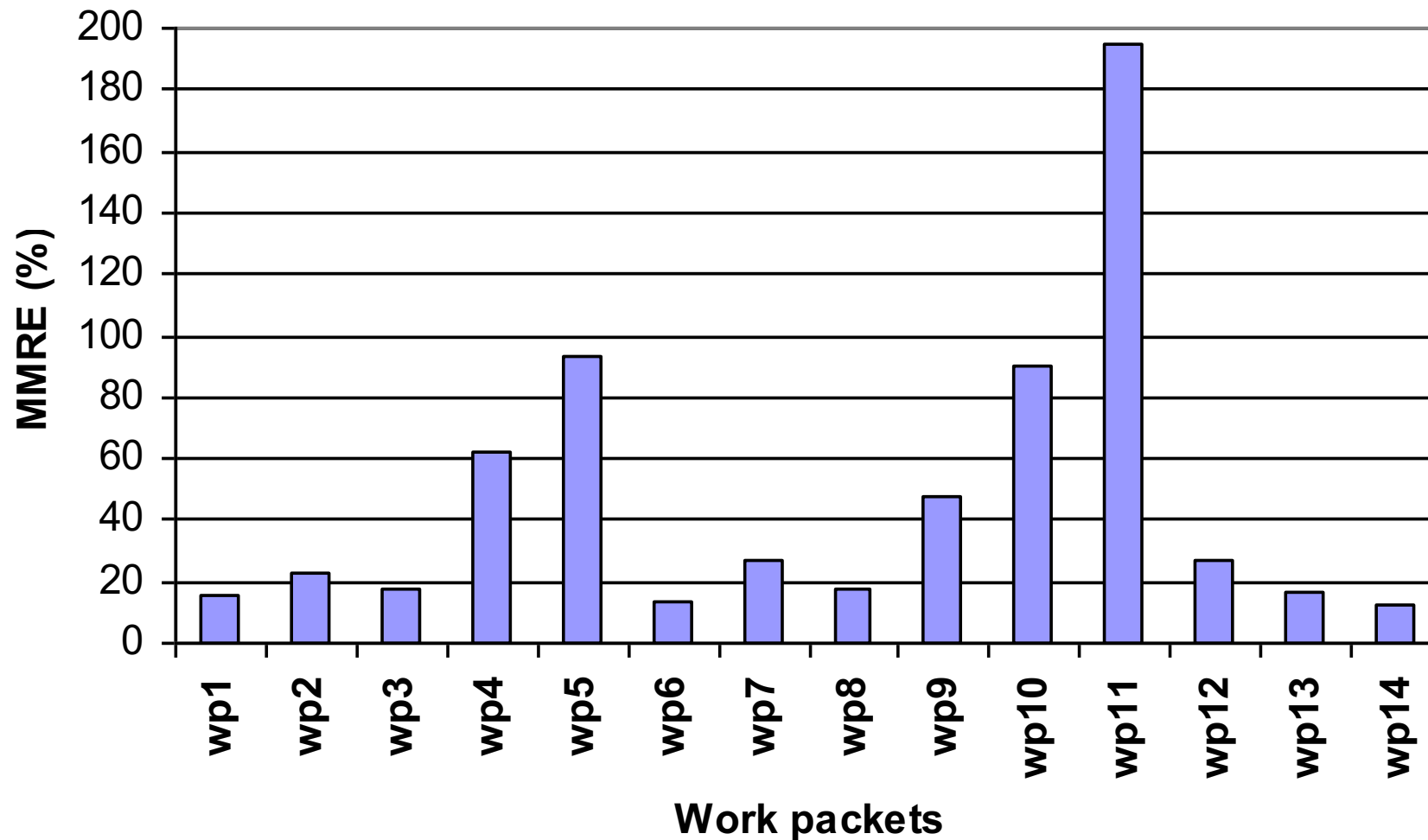


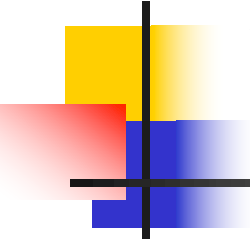
Model Validation

- Leave-one-out cross-validation:
 - Of the 14 WP, one was used to test the model
 - On the remaining WP, build the model
 - The process was repeated a high # of times (100)
 - For each WP, MREs for each experiment and their mean value (MMRE) were computed

$$MRE_k = 100 \frac{|\hat{y}_k - y_k|}{y_k}$$

MMRE for each WP





Varying sample size

% of SC	Avg. MMRE (%)	PRED₂₅ (%)	PRED₅₀ (%)
10	47.1	50.0	71.4
20	46.6	57.1	78.6
30	41.9	57.1	78.6
40	40.7	57.1	78.6